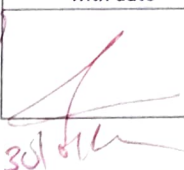


Name: Amey Badgajare Class: 60C 08 Batch: A
Subject: IDS Roll No.: 7 Exp. No.: 1

Name of the Experiment: _____

Performed on: _____

Submitted on: _____

Marks	Teacher's Signature with date
10	

Q1 An algorithm is a finite sequence of well defined, & step-by-step instruction used to solved a specific problem or perform a computation.

characteristics

1. Input : An algorithm takes zero or more input

2. Output : It produces at least one output

3. Definiteness : Each step is clear, precise, unambiguous

4. Finiteness : The algorithm terminates after a finite number of step.

Q2 A data structure is a way of organization storing, and managing data in memory that it can be accessed and modified efficiently.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐

Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐

Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

An abstract data type (ADT) is a logical description of data type that specifies the operations that can be performed on it without specifying how these operations are implemented.

Examples.

- Data structures : Arrays
- ADT : Stack (operation : push, pop, peek)

- Q3.
1. Problem Definition : Understanding requirement and constraints
 2. Analysis : Identify input, output, and processing
 3. Algorithm Design : Create step-by-step logic to solve the problem
 4. flowchart / Pseudocode : Represent the algorithm visually or textually.
 5. Coding : Convert algorithm into a program using a programming language.
 6. Testing & Debugging : Check correctness & removes errors
 7. Documents & maintenance : Improves & maintain the program

Role of Algorithm

The algorithm acts as a blueprint of the solution. It ensures correctness, efficiency & makes coding easier and error-free.

Q4. Linear Data Structure :

Data elements are arranged sequentially

- Array
- Stack
- Queue
- Linked list.

Non Linear Data Structure :

Data elements are arranged in a hierarchical or interconnected manner

Example :-

- Tree
- Graph

Q5

Algorithm

Program

Step-by-step logic

Written code

Language independent

Language dependent

Design Level

Implementation level

No syntax rules

Follows syntax rule.

Role in SDLC :

- Algorithm : Used in design phase to plan the solution
- Program : Used in implementation phase to execute the solution on a computer

Q6] Relationship.

- Data in raw information
- Data structure organize the data
- Algorithm process the data using data structures

Justification:

The efficiency of an algorithm depends on the data structures used

- Searching in an array $\rightarrow$ slower
- Searching in a binary search tree - faster

Conclusion

Choosing the right data structure improves time and space efficiency of Algorithm